

3000 BC – 1776

Early Management

1911 – 1947

Classical Approach

Late 1700s – 1950s

Behavioral Approach

1940s – 1950s

Quantitative Approach

1960s – present

Contemporary Approaches

QUANTITATIVE Approach

Although passengers bumping into each other when trying to find their seats on an airplane can be a mild annoyance for them, it's a bigger problem for airlines because lines get backed up, slowing down how quickly the plane can get back in the air. Based on research in space-time geometry, one airline innovated a unique boarding process called "reverse pyramid" that has saved at least two minutes in boarding time.⁹ This is an example of the **quantitative approach**, which is the use of quantitative techniques to improve decision making. This approach also is known as *management science*.

The quantitative approach evolved from mathematical and statistical solutions developed for military problems during World War II. After the war was over, many of these techniques used for military problems were applied to businesses. For example, one group of military officers, nicknamed the Whiz Kids, joined Ford Motor Company in the mid-1940s and immediately began using statistical methods and quantitative models to improve decision making.

What exactly does the quantitative approach do? It involves applying statistics, optimization models, information models, computer simulations, and other quantitative techniques to management activities. Linear programming, for instance, is a technique that managers use to improve resource allocation decisions; work scheduling can be more efficient as a result of critical-path scheduling analysis; and the economic order quantity model helps managers determine optimal inventory levels. Each of these is an example of quantitative techniques being applied to improve managerial decision making. Another area where quantitative techniques are used frequently is in total quality management.

A quality revolution swept through both the business and public sectors in the 1980s and 1990s.¹⁰ It was inspired by a small group of quality experts, the most famous being W. Edwards Deming (pictured at right) and Joseph M. Juran. The ideas and techniques they advocated in the 1950s had few supporters in the United States but were enthusiastically embraced by Japanese organizations. As Japanese manufacturers began beating US competitors in quality comparisons, however, Western managers soon took a more serious look at Deming's and Juran's ideas, which became the basis for today's quality-management programs.

Total quality management, or **TQM**, is a management philosophy devoted to continual improvement and responding to customer needs and expectations. (See Exhibit MH-6.) The term *customer* includes anyone who interacts with the organization's product or services, internally or externally. It encompasses employees and suppliers, as well as the people who purchase the organization's goods or services. *Continual improvement* isn't possible without accurate measurements, which require statistical techniques that measure every critical variable in the organization's work processes. These measurements are compared against standards to identify and correct problems.

HOW TODAY'S MANAGERS USE THE QUANTITATIVE APPROACH The quantitative approach contributes directly to management decision making in the areas of planning and control. For instance, when managers make budgeting, queuing, scheduling, quality control, and similar decisions, they typically rely on quantitative techniques. Specialized software has made the use of these techniques less intimidating for managers, although many still feel anxious about using them.

MH1.4



This tabulating machine, used in the 1920 United States census, was a predecessor of electronic computers and worked by mechanically reading punch cards with coded information.

Source: Everett Collection Inc/
Alamy Stock Photo



W. Edwards Deming

W. Edwards Deming was one of the founders of the quality revolution.

Source: Richard Drew/AP Images

quantitative approach

The use of quantitative techniques to improve decision making

total quality management (TQM)

A philosophy of management that is driven by continuous improvement and responsiveness to customer needs and expectations

Exhibit MH-6

What Is Quality Management?

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1. **Intense focus on the customer.** The customer includes outsiders who buy the organization's products or services and internal customers who interact with and serve others in the organization.
 2. **Concern for continual improvement.** Quality management is a commitment to never being satisfied. "Very good" is not good enough. Quality can always be improved.
 3. **Process focused.** Quality management focuses on work processes as the quality of goods and services is continually improved.
 4. **Improvement in the quality of everything the organization does.** This relates to the final product, how the organization handles deliveries, how rapidly it responds to complaints, how politely the phones are answered, and the like.
 5. **Accurate measurement.** Quality management uses statistical techniques to measure every critical variable in the organization's operations. These are compared against standards to identify problems, trace them to their roots, and eliminate their causes.
 6. **Empowerment of employees.** Quality management involves the people on the line in the improvement process. Teams are widely used in quality management programs as empowerment vehicles for finding and solving problems.
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